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SarRec: Statistically-guaranteed Augmented Retrieval for Recommendation

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SarRec: Statistically-guaranteed Augmented Retrieval for Recommendation

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Abstract

Recently, Large Language Models with Retrieval-Augmented Generation (RAG) have recently emerged as a powerful paradigm for sequential recommendation. However, existing methods typically retrieve items for each user without any principled mechanism for guaranteeing the reliability of generated recommendations, limiting their trustworthiness. To address this, we introduce **SarRec**: Statistically-guaranteed Augmented Retrieval for Recommendations, a framework that uses a simple retrieval step to provide relevant context and delivers calibrated, uncertainty-aware predictions with formal statistical guarantees. Specifically, SarRec first constructs the user's context set, utilizing a lightweight differentiable retrieval mechanism for identifying relevant context, and then calibrates the LLM's outputs by adapting the conformal prediction mechanism. We further provide a theoretical analysis that establishes an upper bound on the expected risk of recommendation performance metrics. Extensive experiments on multiple datasets from different domains validate the effectiveness of our framework.

CCS Concepts

• Information systems → Data management systems; Data mining; Recommender systems.

Keywords

Recommendation System, Conformal Prediction, Retrieval-Augmented Generation (RAG)

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1 Introduction

Sequential recommendation utilizes the users' historical interactions to model their preferences [19]. In recent years, large language models (LLMs) have shown strong potential in such tasks, particularly within the RAG paradigm. Here, the external contextual information, such as item metadata or user interaction histories, is retrieved and incorporated into the model's input to enhance generation-based predictions [16, 41]. Conventionally, the RAG architectures firstly retrieves a subset of relevant items, and then utilize a generator that conditions on this retrieved context to produce predictions [22, 27]. This decoupled design can be inefficient, as errors from the retriever often propagate to the generator and limit overall performance. To mitigate this, recent works have explored joint optimization of the retriever and generator to improve overall effectiveness [17, 39]. However, a persistent gap remains: these methods rarely incorporate mechanisms to quantify and control prediction uncertainty. Without calibrated confidence guarantees, recommendation sets may vary widely in reliability across users, particularly when interaction histories differ in length or information density. This lack of consistency can lead to incomplete or less trustworthy recommendations, ultimately affecting user satisfaction and long-term engagement [11].

Motivated by this challenge, we aim to develop a framework that delivers statistically reliable recommendations in retrieval-augmented settings. Specifically, instead of solely relying on optimizing retrieval, we aim to design a calibrated predictor that produces recommendation sets with a user-specified confidence level. However, two challenges remain: (1) how can we construct a prediction mechanism that adapts its output to the difficulty or ambiguity of each recommendation context, and (2) how can we quantify uncertainty at the prediction level, ensuring that the model's output includes the true next item with a user-specified level of confidence?

To address these questions, we introduce **SarRec** — Statistically-guaranteed Augmented Recommendation, a two-stage framework that integrates retrieval with statistically calibrated prediction. In SarRec, a differentiable retriever firstly provides relevant candidate items for each user context. Next, we apply a post-hoc calibration

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procedure to the generator's output, using a held-out calibration set to learn a predictive score threshold. This learned threshold creates dynamic recommendation sets for each user such that the risk of missing the relevant items is bounded within a user-specified value. We summarize our contributions as follows:

- We propose **SarRec**, a retrieval-augmented recommendation framework that delivers statistically calibrated recommendation sets. It utilizes a lightweight context-aware differentiable retrieval step, followed by a post-hoc calibration procedure that provides conformal prediction guarantees on inclusion of relevant items in the recommendation set with user-specified confidence.
- Theoretically, we establish statistical guarantees for SarRec in Section 4. We prove that expected utility-based prediction risk at inference remains bounded by a user-defined threshold α , with high probability $1 - \delta$, even with the varying recommendation difficulty in the generator's output due to varying context-specific uncertainty across users.
- Empirically, we conduct extensive experiments across multiple recommendation datasets and various baselines. The results demonstrate that SarRec consistently outperforms the baselines, validating its effectiveness consistent with its theoretical guarantees.

2 Preliminaries

We consider the sequential recommendation task, where the goal is to predict the next item a user is likely to interact with, given their historical interactions. Let $\mathcal{I}$ denote the set of items. For a user u at timestep t , we define their interaction history as a sequence $\mathcal{H}_u = [i_1, i_2, \dots, i_t]$ with each $i_j \in \mathcal{I}$. The task is, therefore, to predict the next item $i_{t+1} \in \mathcal{I}$ that the user will engage with, conditioned on $\mathcal{H}_u$.

We adopt the RAG paradigm: a retriever first selects candidate items relevant to the user's history, and a generator i.e. an LLM, uses these items as additional context to generate predictions.

Formally, the retriever R^ψ , parameterized by ψ , assigns a relevance score:

$$s_{u,i} = R^\psi(\mathcal{H}_u, i), \quad i \in \mathcal{I}. \quad (1)$$

In conventional RAG frameworks, a fixed number of top-ranked items is selected based on Equation (1). Specifically:

$$d(\mathcal{H}_u; k) = \text{Top-}k(s_{u,i} \mid i \in \mathcal{I}), \quad (2)$$

where Top- k returns the indices of the k highest-scoring items.

The resulting set $\mathbf{d} = [d_1, \dots, d_k] \in \mathcal{I}^k$ is inserted into a prompt:

$$\begin{aligned} P(\mathcal{H}_u, \mathbf{d}) = & \text{"User history: } i_1, \dots, i_t. \\ & \text{Candidates: } d_1, \dots, d_k. \text{ What's next?"} \end{aligned} \quad (3)$$

The prompt generated in Equation (3) is then passed to the generator, yielding a probability distribution:

$$\pi_\theta(i_{t+1} \mid \mathcal{H}_u, \mathbf{d}), \quad y \in \mathcal{I}, \quad (4)$$

which represents the model's confidence that item i_{t+1} will be the next interaction.

While retrieval-augmented generation incorporates context, it neglects variability in user interaction histories and the associated uncertainty in predictions. To address this limitation, we couple retrieval with a statistically calibrated prediction stage that adapts

the final recommendation set to meet user-specified confidence guarantees, as detailed next.

3 Proposed Framework

In this section, we introduce our framework, SarRec, which incorporates a retrieval step to identify relevant user context and then provides statistical guarantees on prediction performance across diverse user contexts. The overall structure of our framework is illustrated in Figure 1.

To begin with, **SarRec** uses a lightweight, threshold-based retrieval step that filters the item pool to those most relevant for a given user. It does so by retaining only items whose relevance scores exceed a learnable threshold parameter, i.e. $\kappa \in \mathbb{R}$. Specifically:

$$d(\mathcal{H}_u; \kappa) = \{i \in \mathcal{I} \mid s_{u,i} \geq \kappa\}. \quad (5)$$

To enable end-to-end training, the binary inclusion indicator $\mathbb{I}\{s_{u,i} \geq \kappa\}$ in the above equation is replaced with a smooth sigmoid-based relaxation that assigns each candidate item a soft retrieval weight $\tilde{w}_i \in [\epsilon, 1]$:

$$\tilde{w}_i = \max\left\{\sigma\left(\frac{s_{u,i} - \kappa}{\gamma}\right), \epsilon\right\}, \quad \forall i \in \mathcal{I}, \quad (6)$$

where $\gamma > 0$ is a temperature hyperparameter, $\sigma(\cdot)$ is the sigmoid function, and $\epsilon \ll 1$ ensures differentiability for items selection.

In practice, to ensure that only semantically relevant candidates influence the prediction, items with $\tilde{w}_i \approx \epsilon$ are excluded from the generator input, and the remaining items are used to construct the prompt $P(\mathcal{H}_u, \tilde{\mathbf{w}})$ as defined in Equation (3), encoding both user history and retrieved context. The generator then produces the predictive distribution:

$$\pi_\theta(i_{t+1} \mid \mathcal{H}_u, \tilde{\mathbf{w}}) = \text{Generator}_\theta(P(\mathcal{H}_u, \tilde{\mathbf{w}})). \quad (7)$$

The retriever and generator are then trained jointly using the objective:

$$\mathcal{L}_{\text{train}}(\psi, \theta, \kappa) = \sum_u \sum_{t=1}^{T_u} \left[-\log \pi_\theta(y \mid \mathcal{H}_u, \tilde{\mathbf{w}}) + \eta \sum_{i \in \mathcal{I}} \tilde{w}_i \right], \quad (8)$$

where y is the ground-truth next item and $\eta > 0$ regularizes the size of the retrieved set to discourage overly large candidate pools.

Meanwhile, another tough challenge is ensuring that the ranked list returned by the jointly trained retriever and generator consistently satisfies the desired confidence level. Even with a focused set of retrieved items, prediction uncertainty can vary substantially across users and contexts.

To solve this challenge, we draw inspiration from conformal prediction [2, 6] and extend its coverage guarantees to recommender systems by controlling risk with respect to utility-based performance metrics. Specifically, we construct a prediction set from a score threshold $\lambda \in \Delta \subset \mathbb{R}$, where $\Delta := [0, 1]$ denotes the admissible range of threshold values. Given a user history $\mathcal{H}_u$, the prediction set is defined as

$$C_\lambda(\mathcal{H}_u) = \{i_{t+1} \in \mathcal{I} : \pi_\theta(i_{t+1} \mid \mathcal{H}_u, d(\mathcal{H}_u; \kappa)) \geq \lambda\}, \quad (9)$$

where $\pi_\theta(\cdot)$ is the generator's probability and $d(\mathcal{H}_u; \kappa)$ is the retrieved context from the previous stage. For brevity, we write $C_\lambda := C_\lambda(\mathcal{H}_u)$.

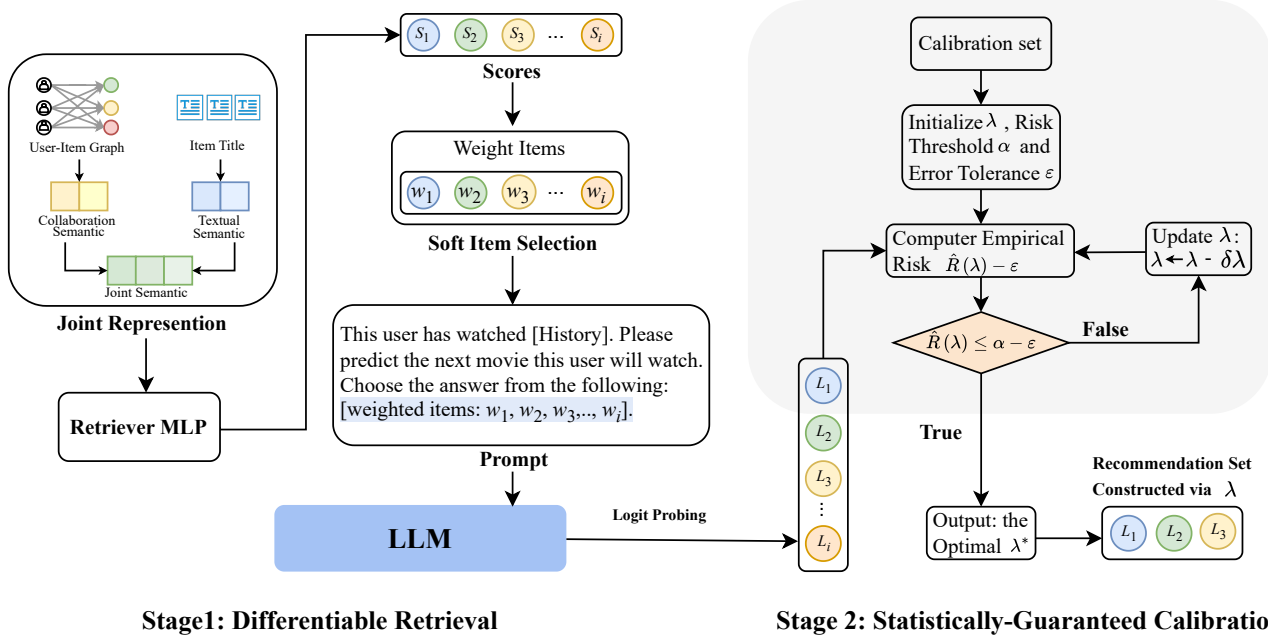


Figure 1: Overview of the SarRec framework. Firstly, it retrieves contextually relevant item, then it applies conformal prediction for statistically guaranteed calibration, generating dynamic recommendation sets under user-defined risk constraints.

Our goal is to select λ such that the prediction sets satisfy a user-specified error tolerance $\alpha \in (0, 1)$. Specifically:

$$\mathcal{R}(C_\lambda) \leq \alpha. \quad (10)$$

Here, the risk is defined as:

$$\mathcal{R}(C_\lambda) = \mathbb{E}_{(\mathcal{H}, y) \sim \mathcal{P}} [\mathcal{L}(y, C_\lambda)], \quad (11)$$

with a bounded utility-based loss

$$\mathcal{L}(y, C_\lambda) = 1 - U_M(y, C_\lambda). \quad (12)$$

The function $U_M(y, C) \in [0, 1]$ is a task-specific recommendation metric (e.g., hit rate, recall, NDCG etc.) that measures the quality of set C with the relevant item as y .

In practice, we estimate the risk on a held-out calibration set $\mathcal{D}_{\text{cal}}$,

$$\hat{\mathcal{R}}(C_\lambda) = \frac{1}{|\mathcal{D}_{\text{cal}}|} \sum_{(\mathcal{H}, y) \in \mathcal{D}_{\text{cal}}} [1 - U_M(y, C_\lambda)], \quad (13)$$

This completes the modeling of our calibrated prediction framework. To output dynamic prediction sets that satisfy the user-defined risk constraint α , we introduce a simple yet effective algorithm which searches over $\lambda \in \Delta$ to find the largest threshold λ^* satisfying the empirical risk bound $\hat{\mathcal{R}}(C_\lambda) \leq \alpha$ on the calibration set. The SarRec framework is depicted in Figure 1 and the algorithm is summarized in Algorithm 1.

Prediction Set Construction: After obtaining the calibrated threshold λ^* from Algorithm 1, we use it to construct prediction sets on the test set $\mathcal{D}_{\text{test}}$. For each user u , we retrieve context items using the learned retriever threshold κ and generate prediction sets by including all items whose generation scores exceed λ^* . These

Algorithm 1: The SarRec Algorithm

Input: Training set $\mathcal{D}_{\text{train}}$, Calibration set $\mathcal{D}_{\text{cal}}$, risk target α , regularisation weight η , temperature γ , small constant ϵ , threshold grid Δ

Output: trained retriever R^ψ , trained generator θ , calibrated threshold λ^*

- 1 **Differentiable Retrieval** Randomly initialize ψ , θ , and κ ;
 - 2 **foreach** mini-batch $\mathcal{B} \subset \mathcal{D}_{\text{train}}$ **do**
 - 3 **foreach** $(\mathcal{H}_u, i_{t+1}) \in \mathcal{B}$ **do**
 - 4 Compute scores $s_{u,i}$ and soft weights $\tilde{w}_i$ using Eq. (6);
 - 5 Build prompt $P(\mathcal{H}_u, \tilde{w})$ and evaluate $\log \pi_\theta(i_{t+1} | \mathcal{H}_u, \tilde{w})$;
 - 6 Accumulate loss $-\log \pi_\theta(y | \mathcal{H}_u, \tilde{w}) + \eta \sum_i \tilde{w}_i$ Eq. (8);
 - 7 update ψ , θ , and κ with optimizer step;
 - 8 **Statistically-Guaranteed Calibration** $\lambda^* \leftarrow \text{None}$;
 - 9 **for** $\lambda \in \Delta$ **do**
 - 10 compute Empirical risk $\hat{\mathcal{R}}(\lambda)$ on $\mathcal{D}_{\text{cal}}$ Eq. (13);
 - 11 **if** $\hat{\mathcal{R}}(\lambda) \leq \alpha$ **then**
 - 12 $\lambda^* \leftarrow \lambda$; **break**;
 - 13 **return** $R^\psi, \theta, \lambda^*$
-

dynamic prediction sets are guaranteed to satisfy the target utility-based risk constraint $\mathcal{R}(C_{\lambda^*}) \leq \alpha$ with high confidence.

4 Theoretical Analysis

In the previous sections we showed how SarRec calibrates a prediction threshold λ^* on a held-out calibration set $\mathcal{D}_{\text{calib}}$. At test time, λ^* together with the learned retrieval threshold κ produces a *dynamic* recommendation set for each user. However, it remains to be seen whether the threshold learned using empirical risk guarantees that the expected risk remains below the user-specified tolerance α . In this section, we utilize a KL-divergence tail inequality combined with a union bound over the finite grid Λ of candidate thresholds to establish that the expected risk deviation remains bounded with high probability. This result is depicted in the following theorem:

THEOREM 4.1 (RETRIEVER-AWARE EXPECTED-RISK GUARANTEE). *Let $\Lambda \subset \mathbb{R}$ be a finite threshold grid. Given a calibration set $\mathcal{D}_{\text{calib}} = \{(\mathcal{H}_i, y_i)\}_{i=1}^n$ and prediction sets $C_\lambda(\mathcal{H}_i)$ defined in Equation (9), assume the utility loss satisfies $L(y_i, C_\lambda(\mathcal{H}_i)) \in [0, 1]$ for all $\lambda \in \Lambda$. Denote expected and empirical risks by $R(\lambda)$ and $\hat{R}(\lambda)$ as in Equations (11) and (13), and choose $\lambda^* := \max\{\lambda \in \Lambda \mid \hat{R}(\lambda) \leq \alpha\}$. Given $\delta := \frac{\log(2|\Lambda|/\eta)}{n}$, then, for any $\eta \in (0, 1)$, with probability at least $1 - \eta$,*

$$R(\lambda^*) \leq \alpha + \varepsilon_{\text{KL}}(\alpha, \delta),$$

where $\varepsilon_{\text{KL}}(\alpha, \delta)$ is the smallest $\varepsilon \geq 0$ such that $D_{\text{KL}}(\alpha + \varepsilon \parallel \alpha) \geq \delta$, and $D_{\text{KL}}(\cdot)$ denotes the binary KL divergence.

PROOF. We aim to control the upper deviation of the expected risk from the empirical estimate, i.e., $R(\lambda) - \hat{R}(\lambda)$, uniformly over all thresholds $\lambda \in \Lambda$, under the influence of the learned retriever threshold κ , which is held fixed during calibration.

Let $Z_i(\lambda) := \mathcal{L}(y_i, C_\lambda(\mathcal{H}_i)) \in [0, 1]$, such that

$$\hat{R}(\lambda) := \frac{1}{n} \sum_{j=1}^n Z_j(\lambda) =: \hat{\mu}, \quad R(\lambda) := \mathbb{E}[Z_i(\lambda)] =: \mu.$$

To control the deviation $\hat{\mu} - \mu$, we use an exponential moment bound on the centered moment generating function (MGF). For any $t > 0$, the MGF of a bounded random variable $Z_i \in [0, 1]$ is defined as:

$$\mathbb{E}[e^{t(Z_i - \mu)}] \leq \exp(\mu(e^t - t - 1)). \quad (\text{i})$$

Let $S := \sum_{i=1}^n (Z_i - \mu)$. Then:

$$\mathbb{P}(\hat{\mu} - \mu \geq \varepsilon) = \mathbb{P}\left(\frac{S}{n} \geq \varepsilon\right). \quad (\text{ii})$$

To bound this, we apply Markov's inequality to the non-negative random variable e^{tS} , which gives:

$$\mathbb{P}(S \geq n\varepsilon) = \mathbb{P}(e^{tS} \geq e^{tn\varepsilon}) \leq \frac{\mathbb{E}[e^{tS}]}{e^{tn\varepsilon}}. \quad (\text{iii})$$

From Eq. (i), we know:

$$\mathbb{E}[e^{tS}] \leq \exp(n\mu(e^t - t - 1)).$$

Substituting this into Eq. (iii) gives:

$$\mathbb{P}(\hat{\mu} - \mu \geq \varepsilon) \leq \exp(-tn\varepsilon + n\mu(e^t - t - 1)). \quad (\text{iv})$$

We will minimize this bound over all $t > 0$ to tighten the concentration inequality. To find the tightest bound, we minimize the exponent with respect to $t > 0$. Let us define the exponent as a function:

$$\psi(t) := -tn\varepsilon + n\mu(e^t - t - 1).$$

Differentiating with respect to t , we obtain:

$$\frac{d\psi}{dt} = -n\varepsilon + n\mu(e^t - 1).$$

Setting the derivative to zero:

$$-n\varepsilon + n\mu(e^t - 1) = 0 \quad \Rightarrow \quad e^t = 1 + \frac{\varepsilon}{\mu} \quad \Rightarrow \quad t^* = \log\left(1 + \frac{\varepsilon}{\mu}\right). \quad (\text{v})$$

Substituting $t = t^*$ back into the bound Eq. (iv) and simplifying we get:

$$\begin{aligned} \mathbb{P}(\hat{\mu} - \mu \geq \varepsilon) &\leq \exp\left(-n\varepsilon \log\left(1 + \frac{\varepsilon}{\mu}\right) + n\mu\left(\left(1 + \frac{\varepsilon}{\mu}\right) - \log\left(1 + \frac{\varepsilon}{\mu}\right) - 1\right)\right) \\ &= \exp\left(-n\left[(\mu + \varepsilon) \log\left(1 + \frac{\varepsilon}{\mu}\right) - \varepsilon\right]\right). \end{aligned} \quad (\text{vi})$$

We now take $p = \mu + \varepsilon$, $q = \mu$, and using the definition of KL divergence, we have:

$$D_{\text{KL}}(\mu + \varepsilon \parallel \mu) = (\mu + \varepsilon) \log\left(1 + \frac{\varepsilon}{\mu}\right) + (1 - \mu - \varepsilon) \log\left(\frac{1 - \mu - \varepsilon}{1 - \mu}\right). \quad (\text{vii})$$

To simplify further, we may also invoke the inequality:

$$D_{\text{KL}}(p \parallel q) \geq p \log\left(\frac{p}{q}\right) - (p - q), \quad \text{for } 0 < q < p < 1.$$

Using this inequality and applying it to Eq. (vii), we obtain:

$$\mathbb{P}(\hat{\mu} - \mu \geq \varepsilon) \leq \exp(-n \cdot D_{\text{KL}}(\mu + \varepsilon \parallel \mu)). \quad (\text{ix})$$

We now apply a union bound over the finite set Λ . For any fixed $\varepsilon > 0$,

$$\mathbb{P}(\exists \lambda \in \Lambda : \hat{R}(\lambda) - R(\lambda) > \varepsilon) \leq |\Lambda| \cdot \exp(-n D_{\text{KL}}(\mu + \varepsilon \parallel \mu)). \quad (\text{xiii})$$

To ensure that the failure probability is at most η , we require:

$$|\Lambda| \cdot \exp(-n D_{\text{KL}}(\mu + \varepsilon \parallel \mu)) \leq \eta. \quad (\text{xiv})$$

Taking logarithms and rearranging terms, we get:

$$D_{\text{KL}}(\mu + \varepsilon \parallel \mu) \geq \frac{\log(2|\Lambda|/\eta)}{n} =: \delta. \quad (\text{xv})$$

We now define the inverse mapping of this inequality, i.e., the smallest $\varepsilon > 0$ such that the KL divergence condition holds:

$$\varepsilon_{\text{KL}}(\mu, \delta) := \inf\{\varepsilon > 0 \mid D_{\text{KL}}(\mu + \varepsilon \parallel \mu) \geq \delta\}. \quad (\text{xvi})$$

This defines the minimum upward deviation in the empirical mean that guarantees the expected risk remains below the target level with high probability.

Specifically, with probability at least $1 - \eta$, we have for all $\lambda \in \Lambda$,

$$R(\lambda) \leq \hat{R}(\lambda) + \varepsilon_{\text{KL}}(\hat{R}(\lambda), \delta), \quad (\text{xvi})$$

We know that λ^* is the largest threshold such that $\hat{R}(\lambda^*) \leq \alpha$. Then from Eq. (xvi), it follows:

$$R(\lambda^*) \leq \hat{R}(\lambda^*) + \varepsilon_{\text{KL}}(\alpha, \delta) \leq \alpha + \varepsilon_{\text{KL}}(\alpha, \delta)$$

Hence Proved. $\square$

Remark. Theorem 4.1 provides a uniform high-probability bound on the deviation between expected and empirical risk under retrieval-aware calibration. The term $\varepsilon_{\text{KL}}(\alpha, \delta)$ captures the minimal inflation above α needed to guarantee this control. As the calibration size n increases, $\delta = \log(2|\Lambda|/\eta)/n$ vanishes at rate $O(1/n)$, driving $\varepsilon_{\text{KL}}(\alpha, \delta) \rightarrow 0$. Hence, the expected risk of λ^* converges to the empirical threshold α , ensuring asymptotically sharp guarantees.

5 Experiments and Results

In this section, we perform a comprehensive set of experiments to assess the effectiveness and robustness of the proposed SarRec framework. Specifically, the experiments are designed to investigate the following research questions: **RQ1** How does SarRec perform compared to conventional recommender systems and state-of-the-art RAG based methods in sequential recommendation scenarios? **RQ2** How do the key components of SarRec (i.e., differentiable retriever and conformal risk calibration) individually influence its recommendation performance? **RQ3** How the risk threshold α influences the performance and the average recommendation set sizes? **RQ4** How efficient is the proposed calibration step in terms of additional computational overhead during inference?

Table 1: Dataset Statistics

Dataset	LastFM	Steam	MovieLens-100K
Users	1,220	11,938	943
Items	4,606	3,581	1,682
Interactions	73,510	274,726	100,000

5.1 Datasets

To validate the effectiveness of the proposed SarRec framework, we conduct experiments on three benchmark datasets. (1) **LastFM** [9]: this dataset contains detailed listening histories from the Last.fm online music service, with each record linked to the corresponding artist name. (2) **Steam** [25]: this dataset captures user interactions and textual reviews from the Steam gaming platform, together with associated game titles. (3) **MovieLens-100K** [18]: this benchmark dataset comprises 100,000 user ratings on movies, along with the corresponding movie titles. To ensure data quality and computational feasibility, we filter out users with fewer than 10 reviews in the Steam dataset. We then partition each dataset into training, calibration, and test sets using a 6:3:1 ratio to avoid information leakage. Table 1 summarizes the statistics of all datasets.

5.2 Implementation Details

Our experiments utilize Llama-3.2-3B [36] as the vanilla model, which is adapted through LoRA fine-tuning. We set the LoRA rank to 8, the scaling factor to 16, and a dropout of 0.1, applying these only to the query and value projection layers within the attention blocks. Training is carried out on a single NVIDIA A100 GPU using the AdamW optimizer with a batch size of 128 and an initial learning rate of 2×10^{-5} , for up to five epochs. For the baseline recommenders, we adopt settings consistent with common practice [3]: embeddings of size 64, three hidden layers, and Adam optimization

with a learning rate of 1×10^{-3} and batch size of 256. Different loss functions are applied to each model, namely Bayesian Personalized Ranking for SASRec, cross-entropy for GRU4Rec, and max-margin ranking for Caser. All baseline models are trained for 50 epochs, with early stopping determined by performance on the validation split.

5.3 Baselines

We compare SarRec with three categories of sequential recommenders. Traditional models include: **GRU4Rec** [20] utilizes recurrent neural networks with session-parallel training and ranking-based losses to capture sequential transitions in user interactions; **Caser** [35] employs convolutional neural networks with both horizontal and vertical filters to extract local and global sequential features for recommendation; **SASRec** [26] adopts the Transformer architecture with self-attention mechanisms to model user preference evolution over long sequences. LLM-based recommenders include: **Llama-3** [36] leverages Meta’s large language model with instruction fine-tuning to directly generate recommendation scores through natural language understanding; **MoRec** [21] introduces a mixture-of-experts framework that combines recommendation-specific and language-specific experts using efficient parameter adaptation techniques; **TALLRec** [5] develops a dual-phase training strategy combining general prompt learning with task-specific optimization for enhanced recommendation performance. Retrieval-augmented approaches consist of: **RALLRec** [16] integrates a bi-encoder retrieval system with generative language models, using retrieved candidates to inform the generation process; **ReLLa** [41] combines dense retrieval mechanisms with large language models to support continuous learning from evolving user preferences and item characteristics.

5.4 Evaluation Metrics

We evaluate performance using four ranking metrics, each normalized to $[0, 1]$ and instantiated as the utility function $U_M(y, C_\lambda)$ in Equation (12): **Recall** measures the proportion of relevant items recovered in the prediction set; **Precision** captures the fraction of recommended items that are relevant; **Mean Reciprocal Rank (MRR)** reflects the reciprocal of the rank position of the first relevant item, rewarding earlier hits; and **Normalized Discounted Cumulative Gain (NDCG)** accounts for both the relevance and the rank position of all relevant items using logarithmic discounting. Unlike fixed- K recommenders, SarRec employs an adaptive- K strategy, where each user’s candidate set size is determined by the retrieval module and a learned threshold λ . Final scores are averaged over users using their individualized set sizes, while fixed- K baselines are evaluated with K set to SarRec’s mean set size for fair comparison.

5.5 Overall Performance (RQ1)

We compare SarRec with conventional sequential recommenders, LLM-based recommenders, and retrieval-augmented recommenders, as detailed in Section 5.3. The results are presented in Table 2. From these results, we make the following observations:

- The SarRec framework consistently achieves the highest performance among all baseline methods across all metrics, validating

Table 2: The results of SarRec compared with traditional sequential recommender models, LLM-based methods, and retrieval-augmented recommender models. Bold and underlined indicate the best and the second-best performance.

Model		LastFM				Steam				MovieLens-100k			
		Recall	Prec.	MRR	NDCG	Recall	Prec.	MRR	NDCG	Recall	Prec.	MRR	NDCG
Traditional	GRU4Rec	0.40	0.25	0.36	0.46	0.42	0.21	0.31	0.44	0.38	0.17	0.28	0.36
	Caser	0.43	0.27	0.39	0.47	0.44	0.23	0.34	0.45	0.39	0.18	0.30	0.37
	SASRec	0.47	0.28	0.42	0.51	0.47	0.24	0.36	0.48	0.42	0.20	0.32	0.41
LLM-based	Llama-3	0.62	0.28	0.45	0.55	0.61	0.26	0.47	0.52	0.57	0.24	0.42	0.49
	MoERec	0.68	0.29	0.51	0.62	0.66	0.28	0.48	0.56	0.62	0.25	0.46	0.53
	TALLRec	0.75	0.29	0.55	<u>0.65</u>	0.71	0.29	0.50	0.60	0.70	0.26	0.51	0.59
Retrieval-augmented	RALLRec	<u>0.78</u>	<u>0.34</u>	<u>0.56</u>	0.64	<u>0.75</u>	<u>0.32</u>	0.53	<u>0.62</u>	<u>0.73</u>	<u>0.28</u>	<u>0.53</u>	0.58
	ReLLa	0.76	0.32	0.55	0.62	0.73	0.29	<u>0.54</u>	0.61	0.71	0.25	0.51	<u>0.60</u>
	SarRec (Ours)	0.81	0.36	0.59	0.67	0.78	0.35	0.56	0.64	0.76	0.31	0.54	0.62

Table 3: Ablation study evaluating the impact of Differentiable(Diff.) retriever and calibration component on SarRec across LastFM, Steam, and MovieLens-100k datasets. Bold values indicate the best performance.

Model Variant		LastFM				Steam				MovieLens-100k			
		Recall	Precision	MRR	NDCG	Recall	Precision	MRR	NDCG	Recall	Precision	MRR	NDCG
w/o Diff. Retrieval		0.72	0.30	0.54	0.59	0.71	0.29	0.52	0.57	0.69	0.28	0.51	0.57
w/o Calibration		0.79	0.35	0.57	0.64	0.76	0.34	0.54	0.63	0.74	0.29	0.53	0.60
SarRec		0.81	0.36	0.59	0.67	0.78	0.35	0.56	0.64	0.76	0.31	0.54	0.62

the effectiveness of our conformal calibration strategy in delivering accurate and reliable recommendations.

- LLM-based recommendation systems such as TALLRec and MoRec generally outperform traditional methods by leveraging semantic understanding and contextual reasoning. However, they still fall short of SarRec due to the absence of explicit statistical coverage guarantees.
- Retrieval-augmented approaches like RALLRec and ReLLa benefit from incorporating additional context and typically surpass traditional and purely LLM-based methods. Nevertheless, their performance remains lower than SarRec’s, as they lack the formal reliability control enabled by conformal prediction.
- Overall, SarRec demonstrates robust and consistent performance across datasets of varying densities, maintaining superior recommendation quality in both scenarios and highlighting the broad applicability of our framework.

5.6 Ablation Study (RQ2)

To verify the contribution of the key modules in SarRec, we perform an ablation study with two specific variants: (i) **w/o Differentiable Retriever**, where the retrieval component is removed and the LLM directly generates recommendations based solely on the user’s historical sequence without external retrieval augmentation; (ii) **w/o**

Calibration, where the statistically-guaranteed calibration component is disabled, directly utilizing adaptive retrieval items without conformal thresholding. To ensure fairness, the “w/o Calibration” variant maintains the same average prediction set size as the calibrated variant. Experimental results are presented in Table 3, from which we observe:

- As observed from the “w/o Differentiable Retriever” variant, removing the retrieval component significantly degrades recommendation performance. Without retrieval augmentation, the LLM must generate recommendations based purely on its parametric knowledge and the user’s historical sequence, lacking access to the external item catalog for candidate selection. This forces the model to rely solely on its internal representations of items learned during training, which may be incomplete or outdated. Consequently, the model struggles to identify appropriate candidates from the full item space, leading to substantial performance drops across all evaluation metrics (e.g., Recall, NDCG, etc.).
- Similarly, disabling the calibration module (“w/o Calibration”) also negatively impacts the model’s recommendation quality. This is because while the adaptive retrieval on its own maximises raw recall, without conformal thresholding, it still outputs a fixed-size prediction set for every user. This uniform prediction size

Table 4: Average time (in minutes) taken by SarRec for LastFM, Steam, and MovieLens-100K datasets, with and without the proposed calibration step.

Setting	LastFM	Steam	MovieLens-100K
w/o Calibration	90.45	130.19	80.23
w/ Calibration	92.3	132.31	81.58
Additional overhead	+1.45	+2.12	+1.35

ignores heterogeneity in session ambiguity, i.e., cautious users with sparse histories need larger candidate set to hit the relevant item, whereas long-tailed power users benefit from tighter, high-precision sets. Calibration personalizes the set size by enforcing a utility-based risk bound, thereby restoring a balanced precision–recall trade-off and yielding higher metrics.

- Overall, both modules complement each other. The differentiable retriever provides essential candidate items from the external catalog, enabling the LLM to focus on ranking and selection rather than relying solely on parametric knowledge. Meanwhile, calibration guarantees that the generated recommendation sets maintain statistical reliability without producing overly large or conservative outputs, preserving both precision and user experience. These results confirm that both differentiable retriever and calibration components independently and jointly contribute to the superior performance and robustness of SarRec.

5.7 Parameter Analysis (RQ3)

We further analyze the influence of the risk threshold parameter α on the recommendation performance and prediction set size in SarRec. We systematically vary α from 0.05 to 0.50 with increments of 0.05, and observe its impact on key recommendation metrics (e.g., Recall, NDCG, and MRR) as well as the average prediction set size. To comprehensively illustrate this influence across varying data densities and domains, we conduct our experiments on all three datasets: LastFM, Steam and MovieLens-100k. Experimental results are depicted in Figure 2, where each row corresponds to one dataset, and each plot simultaneously illustrates how recommendation performance and prediction set size change as α varies. From these experiments, we derive several insights:

- Increasing α consistently decreases the average prediction set size across all datasets. For example, in MovieLens-100K, the prediction set size reduces from 7 (at $\alpha = 0.05$) to 3 (at $\alpha = 0.50$). This behaviour aligns with the theoretical result, as increasing α relaxes the risk threshold, allowing the model to generate smaller prediction sets. Intuitively, when the framework accepts a higher tolerance for error, fewer items are required in the set to satisfy the relaxed constraint. This makes the framework especially useful for applications with limited display space (e.g., mobile recommender UIs), where compact yet reliable sets are preferred.
- Also as α increases, recommendation metrics (Recall, MRR, NDCG) decline. For example, Recall on LastFM falls from 0.81 to 0.65 when α is relaxed from 0.05 to 0.50, illustrating the classical trends of conformal analysis. This is because the framework now

needs only to satisfy a looser risk criterion, which comes at the expense of overall ranking accuracy. In practice, this highlights the inherent trade-off between risk tolerance and predictive sharpness: stricter thresholds yield larger sets with higher likelihood of containing the true item, while looser thresholds result in leaner sets but lower ranking fidelity.

- We observe these trends are consistent across MovieLens-100K (dense), Steam, and LastFM (sparse), confirming that the risk performance relationship is dataset-agnostic and that SarRec generalizes well despite differing interaction sparsity and domain semantics. This robustness is important because it demonstrates that conformal calibration in SarRec does not rely on dataset-specific properties but rather stems from principled risk control, making it applicable to a wide range of recommendation domains.
- The observed observations emphasize the practical applicability of our framework as practitioners can flexibly adjust α based on application-specific requirements. For instance, in high-stakes domains such as healthcare or financial recommendations, a small α ensures coverage with strong guarantees, even if larger recommendation sets are produced. Conversely, in entertainment or e-commerce settings where compact, personalized sets are desirable, a moderate α offers a practical balance between coverage and efficiency. Thus, SarRec provides a tunable mechanism to adapt to diverse recommendation scenarios.

5.8 Time Efficiency Analysis (RQ4)

Next, we analyze the computational overhead introduced by the post-hoc calibration stage, using the Llama-3.2-3B fine-tuned model on a single NVIDIA A100 GPU. Specifically, we report the time required to compute the calibrated threshold λ^* on the held-out calibration split for each dataset.

As shown in Table 4, across LastFM, Steam, and MovieLens-100K, the calibration step completes within at most 2.2 minutes. This efficiency arises because calibration consists of a single forward pass with a simple threshold search (no retraining). Consequently, the modest extra time is negligible compared to the performance improvements reported in Table 3. These results confirm that SarRec is not only statistically reliable but also computationally efficient. The minimal overhead demonstrates that our calibration procedure scales effectively to real-world recommendation settings, where both accuracy and efficiency are critical for deployment.

6 Related Works

6.1 LLMs for Recommendation

Sequential recommendation aims to capture the evolving preferences of users by modeling temporal patterns in their interaction sequences, and has been extensively studied through various neural architectures. Early approaches like GRU4Rec [19], Caser [35], SASRec [25], and SIM [42] typically rely on ID embeddings trained on large-scale interaction data and have demonstrated strong performance in capturing collaborative filtering signals from user behavior sequences. Recently, large language models have introduced a transformative paradigm by reformulating user interaction histories as natural language prompts, enabling models to exploit pretrained knowledge alongside advanced text understanding and generation

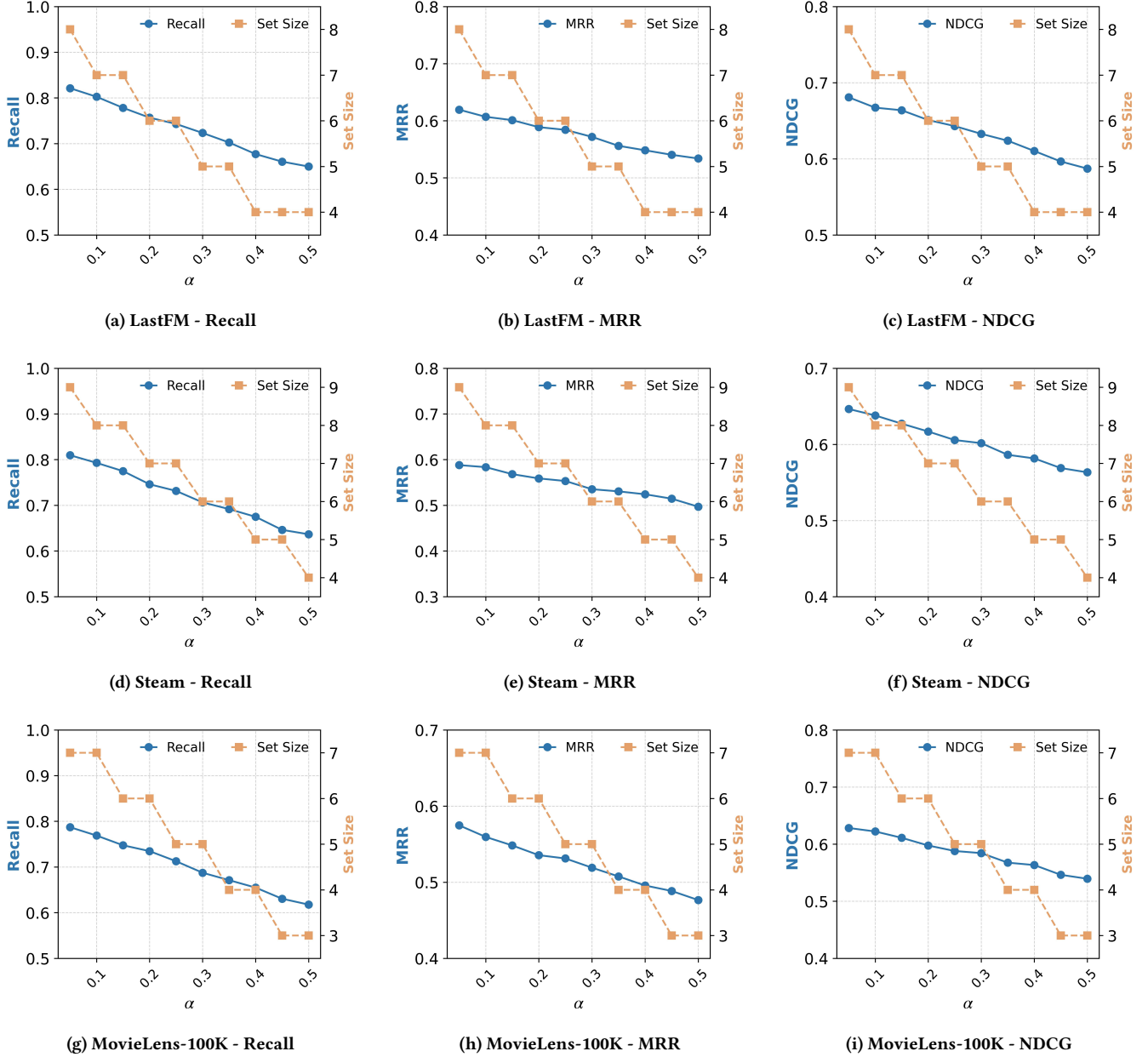


Figure 2: Metric trends with varying α across three datasets. Each row represents one dataset LastFM, Steam, and MovieLens-100K respectively, and columns correspond to performance metrics Recall, MRR, and NDCG.

capabilities for recommendation [12, 30]. This shift allows recommendation systems to leverage the rich semantic understanding and few-shot learning abilities inherent in LLMs, opening new possibilities for handling diverse recommendation scenarios including cold-start users and cross-domain transfer. Early efforts have defined recommendation tasks as language modeling problems: P5 [32] reformulates various recommendation subtasks (e.g. rating prediction, explanation, etc.) into text-to-text prompts, enabling unified training and inference with T5. Similarly, TALLRec [5] applies

instruction tuning on LLMs using handcrafted prompts tailored to CTR prediction. CDPO [28] identifies the severe performance degradation of LLMs on tail items and introduces reembedding and reweighting strategies to address this challenge, while LPO4Rec [29] proposes listwise preference alignment to improve both training efficiency and tail-item recommendation performance. While these LLM-based approaches have demonstrated significant improvements in recommendation quality and semantic understanding,

they primarily focus on accuracy optimization without providing formal reliability guarantees.

6.2 RAG for Recommendation

The emergence of retrieval-augmented generation has further advanced the field by alleviating the memorization burden of LLMs and providing more timely, contextually relevant information for recommendation tasks. Unlike traditional LLM approaches that rely solely on parametric knowledge, RAG-based methods combine the generative capabilities of language models with external retrieval systems to access dynamic and up-to-date item catalogs. ReLLa [41] proposes semantic user behavior retrieval, replacing recency-based item selection with relevance-based candidate retrieval to better match user preferences. RALLRec [37] enhances RAG for recommendation by jointly learning item representations through self-supervised contrastive learning, improving both retrieval effectiveness and generation quality for instruction tuning. To enable end-to-end optimization of retrieval and generation components, Stochastic RAG [40] presents a differentiable optimization framework using Gumbel-top- k sampling techniques. This approach enables gradient flow through discrete retrieval operations, allowing the retrieval module to be optimized jointly with downstream generation objectives. The integration of retrieval and generation has proven particularly effective for sequential recommendation, where the model must balance between exploiting historical user preferences and exploring new, potentially relevant items from large catalogs. However, despite these advances in accuracy and generation quality, existing RAG-based recommendation methods primarily focus on performance optimization without addressing fundamental questions of reliability and uncertainty quantification, leaving open the critical challenge of ensuring statistically well-calibrated recommendations.

6.3 Conformal Prediction for Sequential Recommendation

Conformal Prediction (CP) [14, 31, 33] is a statistical tool that provides a distribution-free framework for quantifying model uncertainty by constructing prediction sets with formal coverage guarantees. It augments its outputs to form a set of labels that, with high probability (e.g., $1 - \alpha$), contains the true outcome [2, 4, 34]. This finite-sample guarantee holds under minimal assumptions and has been highly adopted in high stake domains such as computer vision [2], NLP [8], and large language models [13]. CP achieves this by calibrating a nonconformity score (e.g., rank, error margin, or confidence gap) on a held-out dataset to determine a score threshold that yields valid prediction sets. Some recent works extend CP to settings with broader risk constraints. Adaptive and online variants of CP have been developed to maintain guarantees under distribution shift, such as concept drift in time-series data [1, 15, 38]. These methods recalibrate thresholds dynamically to preserve validity. Some parallel works like conformal risk control [2, 6] generalize CP beyond set coverage, instead of controlling $y \notin C(x)$, these methods control arbitrary loss functionals. This Risk-Controlling Prediction Set (RCPS) framework allows application-specific metrics, such as false-negative rate or utility-based losses, to be bounded while

retaining distribution-free guarantees. Applications of CP in recommendation are emerging but remain limited in scope. Kagita et al. [23, 24] propose conformal extensions to top- N recommendation, using inductive CP to calibrate item scores and guarantee that recommended sets include the relevant item with high probability. More recently, Bisht et al. [7] introduces a fairness-aware conformal framework (CUGF) that enforces group-wise coverage guarantees across user demographics. However, these approaches involve conformal coverage only after scoring on a fixed trained model which inflates their confidence margins, often yielding overly large, conservative prediction sets that clutter the interface and can trigger issues such as ad-blindness or user fatigue [10]. We aim to address these challenges using our framework.

7 Conclusion

In this paper, we introduced SarRec to address the limitations of retriever based LLM recommendation models. SarRec combines a differentiable retrieval mechanism, dynamically adjusting the candidate item set per user context and utilizes a conformal prediction-based calibration method that ensures robust and statistically rigorous recommendation outputs. Through theoretical analysis, we establish formal guarantees on SarRec’s performance, ensuring the expected recommendation risk remains within user-specified thresholds. Extensive experimental evaluations conducted on multiple datasets demonstrated SarRec’s consistent and significant improvements over state-of-the-art baselines across key ranking metrics, confirming its effectiveness and practical value. Ablation studies further emphasize the importance of the differentiable retriever and calibration components in our framework. For future work, we plan to extend SarRec to more complex recommendation tasks and investigate its integration into broader interactive recommendation ecosystems. Overall, our work presents a novel solution for addressing user contextual variability in RAG-based recommendation systems, along with a principled framework that ensures statistical guarantees for performance metrics, marking a practical and forward-looking contribution to future research in this domain.

GenAI Usage Disclosure

Generative AI tools (e.g., ChatGPT) were used strictly for grammar checking and sentence clarity enhancement on human-written text. All model design, data analysis, experiments, and core manuscript content were authored solely by the listed contributors without assistance from text generation tools. No AI-generated figures, tables, or novel content were included.

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